



SGGS INSTITUTE OF ENGINEERING & TECHNOLOGY, VISHNUPURI, NANDED – 431 606

Department of Production Engineering

Academic Year 2025-26

ISE-1 Examination Class : Class: S.Y. Production

Subject: CASTING AND WELDING (PCC-PE-203)

Max Marks: 20

Dt: 19/08/25

11.00-11.45

Note: Solve any TWO questions from the following.

***CO: Course Outcomes., BT-blooms taxonomy**

Q.1	List down the different constituents in flux coating and their role in enhancing welding arc features in SMAW.	10	CO2	BT2
Q.2	What are the different types of patterns? Explain any two of them with their applications.	10	CO1	BT1
Q.3	What is a defect in welding? List down and explain with neat sketches any five defects in welding along with causes.	10	CO2	BT2
Q.4	a. With neat sketches explain the different positions of welding and joints used in welding process. b. Write a note on electrodes used in welding.	10	CO2	BT1



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (Semester I)

Class: SY B.Tech.

Course: Strength of Materials

Course Code: PCC-PE201

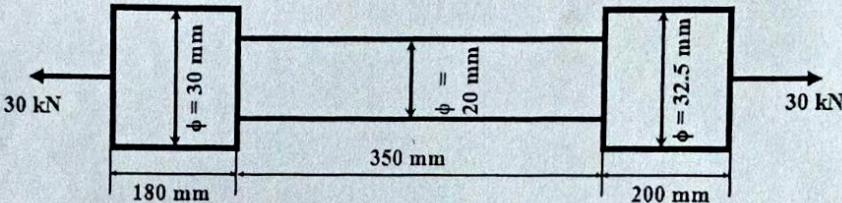
Date: 21.08.2025

Time: 11.00 am to 11.45 am

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	The Following figure shows a stepped rod consisting of three lengths. Find the stresses, extensions in each parts and the total extension of the rod for an axial load of 30 kN. Take $E = 2 \times 10^5 \text{ N/mm}^2$.	10	CO2	BT4
				
Q.2	The principal stresses at a point across two perpendicular planes are 60 MN/m^2 (tensile) and 40 MN/m^2 (tensile). Find the Normal stress, Tangential stress, Resultant stress & its obliquity, maximum shear stress, on a plane at 20° and 60° degrees with the major principal plane. Use analytical and Mohr's circle method.	10	CO3	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:

- CO 1 To differentiate the mechanical behaviour of engineering materials.
- CO 2 To apply the theories of elastic failure.
- CO 3 To determine the shear force and bending moment distribution of the beam and locate the point of contraflexure.
- CO 4 To calculate stresses in beams for various boundary conditions.
- CO 5 To determine the slope and deflection of the beam for various boundary conditions.